



SPECIAL CABLES

**BUILDING RELATIONSHIPS
THROUGH PERFORMANCE**



PRODUCT CATALOGUE

About Us

Special Cables serving the Industry for over three decades as one of the most reputed & reliable Cable Manufacturers

Special cables was incorporated in 1983 as a private limited company and began its business operations in 1985. Starting as a manufacturer of special purpose cables including Screened, Co-axial & Telephone cables, the company has expanded its product range to include Power cables, Control cables, Instrumentation cables, Fire Survival cables, Solar cables, Mining cables, Rubber cables, Composite cables, High Temperature Cables, Railway Power & Signalling Cables, A.B. cables, Overhead conductors, House wires, Multicore Flexible cables etc.

Special Cables is an organisation where the "Customer comes First". Our aim is to excel in all areas of operations through continuous improvement and to create a congenial working environment to ensure long lasting and cordial business relationship with our stakeholders and commitment to environmental issues and society as a whole. Over the years, we have strengthened our brand name and reputation in the market and are proud to be the preferred vendor to many leading organisations and industrial houses.

The company is managed by an enterprising team of highly professional, technically qualified and well experienced personnel having proven expertise in the cable industry and management experts from reputed organisations.

We have the vision, the commitment and the expertise to be the market leader. For us, leadership does not mean the biggest, but the best in terms of quality of our products, timely delivery, value advantage, all round satisfaction and trust for our customers.

Quality Assurance and Timely Delivery continue to be the driving force at Special Cables. In the changing world, our focus remains on supplying high quality products at very competitive prices. Our mission is to continue to remain the preferred supplier to our valued customers and to strive for growth in existing and new markets and build long term relationship with our associates, customers, employees and suppliers.

Production & Quality

Our manufacturing facilities located in Rudrapur, Uttrakhand is focused on achieving high production capacities through new & energy efficient technologies and process improvements. The unit has state-of-the-art manufacturing facilities with machinery from reputed domestic and international manufacturers. The current production floor is spread over 100,000 sq. ft. with continuous power, back-up generators and UPS systems providing uninterrupted power supply. The unit has installed capacity to manufacture wires, cables & conductors upto Rs. 3500 million/annum and provides employment to over 250 workmen.

Quality is an all-encompassing philosophy at Special Cables. We always strive to maintain the highest quality standards and ensure continuous quality improvement, thereby delivering our mission to provide world-class materials and services, to serve our customers better and better.

Our testing laboratories are equipped with modern and computerized testing equipments. Our Quality Management System ensures the quality of product at each stage of manufacture.

Certification

- We are ISO 9001 : 2015, ISO 14001: 2015 and ISO 45001: 2018 certified.
- Approvals from various National and International Organizations.
- Holding valid BIS licenses i.e. IS:694, IS:1554 (Pt-1), IS:7098 (Pt-1 & 2), IS: 9968 (Pt-1) & IS:14225. Type Tests on various cables have been conducted at CPRI, ERDA, NABL and various government labs.



Product Range

POWER CABLES



Application: Power transmission and distribution systems (underground / overhead) in various industries including thermal and nuclear power stations, renewable energy, steel, cement, defence, railways etc.

Types and Sizes: Single core ranging from 4 sq. mm. up to 1000 sq mm. and Multi core (up to 4 core) ranging from 4 sq. mm to 630 sq. mm. with voltage grade 1.1 KV to 3.3 KV.

Conductor: Aluminium – Solid, Stranded (Sector/Circular) or Copper – Solid, Stranded (Sector/Circular) or Flexible

Insulation: PVC- GP/HR/FR/FRLS, PE, XLPE, Zero Halogen

Inner Sheath: PVC- GP/HR/FR/FRLS, PE, Zero Halogen

Armour: Galvanized Steel or Aluminium Round Wire / Flat Strip / Tape

Outer Sheath: PVC- GP/HR/FR/FRLS, PE, Zero Halogen

Specification: Generally confirming to IS: 1554 (Pt-1 & 2), IS: 7098 (Pt-1 & 2), IEC: 60502-1, BS: 5467, BS: 6346

Application: Mainly used for interconnections for control circuits, communication systems, electrical panels, machine tools as well as lighting at lower loads.

Types and Sizes: Typically upto 61 cores from 1.0 sq. mm to 2.5 sq. mm with voltage grade upto 1.1 KV

Conductor: Copper – Solid, Stranded (Circular) or Flexible – Bare or Tinned

Insulation: PVC- GP/HR/FR/FRLS, PE, XLPE, Zero Halogen or EPR Rubber.

Inner Sheath: PVC- GP/HR/FR/FRLS, PE, Zero Halogen or Rubber – FR/HOFR

Armour: Galvanized Steel Round Wire / Flat Strip / Tape or Aluminium Round Wire / Flat Strip or GI / Tinned Copper Braiding

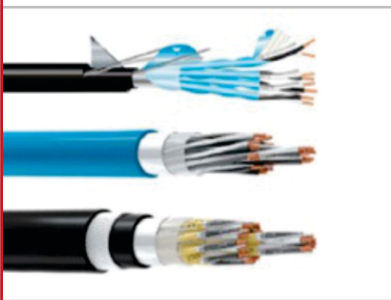
Outer Sheath: PVC - GP/HR/FR/FRLS, PE, Zero Halogen or Rubber- FR/HOFR/TRS

Specification: Generally confirming to IS: 694, IS: 1554 (Pt-I), IS: 7098 (Pt-I), IEC: 60502-1, BS: 5467, BS: 6346.

CONTROL CABLES



INSTRUMENTATION CABLES



Application: Used widely in measurement, control and supervision in process instruments and equipments, various communication and data acquisition systems, computer systems, digital control & measurement systems etc. and are designed to transmit signal without interference.

Types and Sizes: Typically in pair / triad formation (upto 50 Pairs, Triads) from 0.50 sq.mm to 2.5 sq.mm with screening for voltage grade upto 1.1 KV

Conductor: Copper – Solid, Stranded (Circular) or Flexible – Bare or Tinned

Insulation: PVC- GP/HR/FR/FRLS, PE, XLPE, Zero Halogen or EPR Rubber

Screening: Individual & Overall or Overall screening - Polyester Tape, Aluminium Mylar Tape, with Copper Drain Wire, Copper Tape or Tinned Copper Braiding

Inner Sheath: PVC - GP/HR/FR/FRLS, PE, Zero Halogen or Rubber – FR/HOFR

Armour: Galvanized Steel Round Wire / Flat Strip / Tape or GI / Tinned Copper Braiding

Outer Sheath: PVC - GP/HR/FR/FRLS, PE, Zero Halogen or Rubber- FR/HOFR/TRS.

Specification: Generally confirming to IS:1554 (Pt-1), BS: 5308 (Pt-1 & 2), IEC: 189 (Pt 1 & Pt 2), IEC: 60227, VDE: 0815 & 0816, BS EN: 50288

Application: Wide application in nuclear and thermal power stations, defence, windmills, mining, marine & and offshore applications, railways, cranes & reeling applications, lifts and other heavy duty equipment.

Types and Sizes: Single & multi core from 0.50 sq.mm to 630 sq. mm for voltage grade upto 1.1 KV

Conductor: Copper – Solid, Stranded (Circular) or Flexible – Bare or Tinned

Separator: Polyester tape, Fibre Glass Tape, or suitable material tape is applied over conductor

Insulation: General purpose Type IE1, Heat resisting Type IE2/IE3, Flame retardant Type IE4, Silicone Rubber Type IE 5. For identification of cores, coloured insulation / numbered printing / numbered polyester tape / coloured proofed RC tape is used

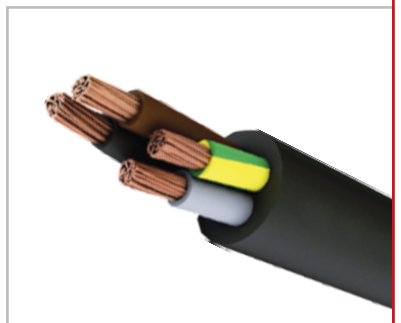
Inner Sheath: General purpose Type SE1/SE2, Heat resisting Type SE3/SE4/SE6 (HOFR), Silicon Rubber Type SE-5

Armour: Copper Wire / Stainless Steel / Fibre Glass Yarn Braiding

Outer Sheath: General purpose Type SE1/SE2, Heat resisting Type SE3/SE4/SE6 (HOFR), Silicon Rubber Type SE-5. Fibre Glass braid with varnish is also available.

Specification: Generally confirming to IS: 9968 (Pt-1), IS 9857, BS: 6500, BS: 7919, BS EN: 60228, BS EN: 60332, IEC: 60502.

RUBBER CABLES



SOLAR CABLES



Application: Solar cables are used for interconnections between solar panels and other electrical components of D.C. system. They are designed to be UV resistant and withstand extreme and demanding weather conditions over long time periods.

Types and Sizes: Typically Single core cables upto 630 sq. mm for voltage grade 1.1 KV to 3.3 KV

Conductor: Aluminium – Stranded (Circular) or Copper – Stranded or Flexible (Circular)

Insulation: XLPE, XLPO, HFFR

Outer Sheath: PVC FR, FRLS, HFFR, HDPE

Specifications: Generally conforming to EN 50618, TUV 2 pfg 1169/08.2007 and designed to meet customer's requirements

Application: Used for very high temperature applications in steel, chemical & metallurgical industry, nuclear power plants, oil exploration & oil rigs, military & defence equipments, aircraft & aerospace vehicle wiring, auto wiring, electrical appliances etc.

Types & Sizes: These cables can start from single core 0.50 sq mm and can be made for various higher sizes depending on customer requirement.

Conductor: Bare / Tinned copper (up to 150° C), Silver plated copper (up to 200° C), Nickel plated conductor (up to 260° C) or Thermocouples depending on customer requirements.

Insulation: PTFE, ETFE, FEP, PFA, PEEK, Silicon and other special compounds to withstand extremely high temperatures.

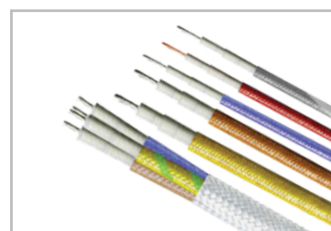
Screening: Individual & Overall or Overall screening - Aluminium Mylar Tape with copper drain wire.

Armouring: Galvanized steel wire / stainless steel wire or Braiding with bare / tinned / nickel plated / silver plated copper wire or fibre glass.

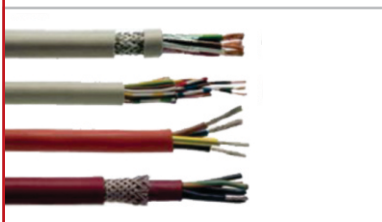
Outer Sheath: FEP, PFA, PEEK, Silicon and other special compounds to withstand extremely high temperatures.

Specification: JSS 51034, JSS 51037, JSS 51038, IS:8130 and designed to meet customer requirements.

HIGH TEMPERATURE CABLES



THERMOCOUPLE EXTENSION/COMPENSATING CABLES



Application: Designed for interconnection between Thermocouple probes and Reference units and used for process temperature measurement and connected to pyrometers for indication and control.

Types and Sizes: KX, JX, TX, EX, SX/RX typically 16, 18 & 20 AWG

Conductor: Chromal Alumel/Iron/Copper Nickel Alloy / Copper Constantan Alloy -Solid / Stranded

Insulation: PVC- HR/FR/FRLS, PE, XLPE, Zero Halogen or EPR Rubber

Screening: Individual & Overall or Overall screening - Polyester Tape, Aluminium Mylar Tape, Copper Tape, Copper Braiding

Inner Sheath: PVC - HR/FR/FRLS, PE, Zero Halogen or Rubber – FR/HOFR/TRS

Armouring: Galvanised Steel Round Wire, Flat Strip, Braiding

Outer Sheath: PVC - HR/FR/FRLS, PE, Zero Halogen or Rubber – FR/HOFR/TRS

Specification: ANSI: MC 96.1, IS: 8784, BS: 4937, IEC: 584, DIN, JIS and designed to meet customer requirements.

Application: Designed to maintain circuit integrity during fire and ensure maximum time for evacuation in case of emergency. Typically used in fire alarm systems, emergency lighting, critical circuits and other safety equipment.

Types and Sizes: Single core up to 1000 sq. mm and multicore up to 400 sq. mm

Conductor: Aluminium – Solid, Stranded (Sector/Circular) or Copper – Solid, Stranded (Sector/Circular) or Flexible - Bare or Tinned

Fire Barrier: Layers of Glass Mica Tape over Conductor & Insulation as per requirement.

Insulation: PVC, XLPE, EPR Rubber with Additional taping or screening if required

Inner Sheath: PVC - FR/FRLS, PE, Zero Halogen or Rubber – FR/HOFR.

Armouring: Galvanised Steel Wire / Strip or Braided with Fibre Glass Tape (if required)

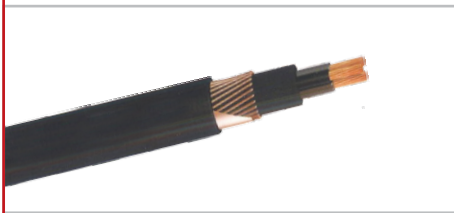
Outer Sheath: PVC - FR/FRLS, PE, Zero Halogen or Rubber – FR/HOFR.

Specification: BS: 7846 or equivalent BS: 6387 Category CWZ, IEC: 60331 and designed to meet customer requirements.

FIRE SURVIVAL CABLES



CONCENTRIC CABLES



Application: Provide final connections to residential properties indoor, outdoor and underground. Also suitable for sub main distribution and within high – rise buildings and street lighting systems.

Types & Sizes: Concentric & Split Concentric cables for voltage upto 1.1 KV.

Conductor: Aluminium, Copper - Solid / Stranded.

Insulation: PVC- HR/FR/FRLS, PE, XLPE, Zero Halogen.

Armour: Concentric Copper / Aluminium Conductor (with insulation) and Copper / Aluminium Tape.

Outer Sheath: PVC- HR/FR/FRLS, Zero Halogen.

Specification: BS: 7870, BS EN 60228, DKS: 1022 and designed to meet customer requirements.

Application: Used in various types of railway applications including distribution & transmission of energy, light switching, track changing and communication etc.

Types and Sizes: Control cables upto 37 Cores for 1.5 sq mm & 2.5 sq mm. and Multicore power cables upto 50 sq.mm. for voltage grade upto 1.1 KV.

Conductor: Aluminium - Solid, Stranded or Copper – Solid Circular Conductor

Insulation: High quality PVC insulation material to meet stringent IRS Properties.

Inner Sheath: High quality PVC sheathing material to meet stringent IRS Properties.

Armour: Galvanised Round Wire / Flat Strip / Double Steel Tape.

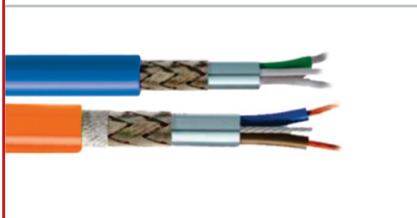
Outer Sheath: High quality PVC sheathing material to meet stringent IRS Properties.

Specifications: Indian Railway Specifications: S-63/2014 & IS: 1554 (Pt-1)

RAILWAY POWER & SIGNALLING CABLES



FOUNDATION FIELDBUS / PROFIBUS CABLES



Application: For Bi-directional communications protocol used for communications amongst field devices and control system. Installed in refining, petrochemicals, power generation, food & beverage, pharmaceuticals and nuclear applications.

Types & Sizes: Typically manufactured in single and multiple pairs with sizes 20 AWG / 18 AWG / 16 AWG / 14 AWG for voltage grade 300/600 V.

Conductor: Bare / Tinned copper (up to 150° C), Silver plated copper (up to 200° C), Nickel plated conductor (up to 260° C).

Insulation: Polyethylene, XLPE, PFA (higher temperatures)

Screening: Individual & Overall or Overall screening - Aluminium Mylar Tape with Tinned Copper Drain Wire, Copper Tape or Braiding

Inner Sheath: PVC- HR/FR/FRLS, Zero Halogen, PA

Armour: Galvanized Steel Round Wire/ Flat Strip/ Steel Wire Braiding, Steel Tape

Outer Sheath: PVC- HR/FR/FRLS, Zero Halogen, PA

Specification: Foundation Fieldbus FF-844H1, EN50288-7, BS: 5308 (Pt-1), IEC 60332, JIS and designed to meet customer requirements.

Application: Reeling & Trailing cables are used in conveyor machinery such as transfer cars, boomstackers, side arm chargers, bulk materials conveyors etc. They find application in almost every industry segment like Steel Mills, Cement plants, Docks, Power Plants, Automobile Industries & Refineries and Petrochemicals.

Type & Sizes: Multicore cables for voltage grade upto 3.3 KV .

Conductor: Copper - Flexible – Bare or Tinned

Insulation: Rubber - HOFR/HR

Screening (if Applicable): Copper wire braiding - Bare or Tinned

Inner Sheath: Rubber - HOFR/HR

Braid: Anti –twisting Textile yarn braiding between inner and outer sheath, if required.

Outer Sheath: Rubber - HOFR/HR

Specification: All cables are designed to meet customer's requirements and generally confirming to IS, IEC, VDE standards.

CRD AND TRAILING CABLES



MINING CABLES



Application: Used across various mining operations in tunnelling, drilling, loading machines, shuttle cars, roof bolters, power feeds, open pit mining, mining pumps etc. They are subject to extremely harsh working conditions with extensive vibrations, impact and tension.

Types and Sizes: Include power, control cables etc. in single and multicore starting from 1.5 sq mm. upto 630 sq.mm. for voltage grade upto 3.3 KV.

Conductor: Aluminium – Stranded (Sector/Circular) or Copper – Stranded (Sector/Circular) or Flexible.

Insulation: PVC-HR/FR/FRLS, XLPE, Zero Halogen or EPR / HEPR Rubber

Inner Sheath: PVC-HR/FR/FRLS, XLPE, Zero Halogen or EPR / HEPR Rubber

Armour: Galvanized Steel or Aluminium Round Wire / Flat Strip. Normally single / double wire armour along with tinned copper to meet conductivity requirements of atleast 50% of phase conductor. Tinned Copper Braiding can also be provided.

Outer Sheath: PVC- HR/FR/FRLS, Zero Halogen or Rubber-TRS/HR/HOFR.

Specification: Generally conforming to IS: 1554 (Pt-1 & 2), IS: 7098 (Pt-1 & 2), IS -14994, IEC: 60502-1, BS: 5467, BS: 6346

Application: These cables are used in defence shipyards and private shipyards in varied applications.

Type & Sizes: Complete range incld. power, control, instrumentation, thermocouple, fire resistant & high temperature cables. Single and multicore starting from 0.50 sq mm. for instrumentation cables upto 1000 sq.mm. for power cables for voltage grade upto 3.3 KV.

Conductor: Copper – Solid. Stranded or Flexible – Bare or Tinned. For Thermocouple – KX, JX, TX, EX, SX/RX

Fire Barrier: Layers of Glass Mica Tape (if required).

Insulation: PVC, HR PVC, FR, FRLS, PE, XLPE, EPR, Zero Halogen or EVA Rubber.

Screening: Individual & Overall or Overall screening - Polyester Tape, Aluminium Mylar Tape, Copper Tape, Braiding with Copper Drain Wire if required.

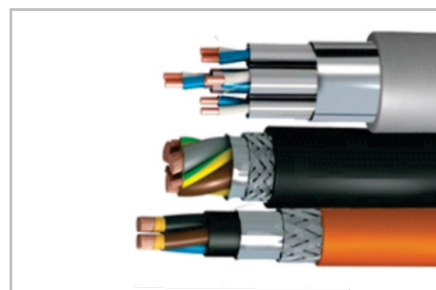
Inner Sheath: SHF1, SHF2 or other polymeric compounds.

Armour: Copper wire – Bare or tinned, GI, SS or Bronze Wire.

Outer Sheath: SHF1, SHF2 or other polymeric compounds.

Specification: IEC 60092, NEK 606 standards and designed to meet customer requirements.

MARINE & OFFSHORE CABLES



HOUSE / BUILDING WIRE



Application: Widely used in residential buildings, commercial buildings, multiplexes, institutions and also in a wide variety of industries in machineries, tools, appliances and control panels.

Types & Sizes: Single core double or triple insulated cables ranging 0.5 to 240 sq mm. for voltage grade 1.1KV.

Conductor: Copper – Solid, Stranded (Circular) or Flexible – Bare or Tinned

Insulation with Dual Extrusion: PVC HR/FR/FRLS, Zero Halogen

Specifications: IS: 694, BS: 6500, IEC 60227 and meeting customer requirements

Application: These cables are used mostly in residential buildings, commercial buildings, multiplexes, institutions and also in a wide variety of industries in machineries, tools, appliances and control panels. Flat cables are used in Submersible pumps.

Type & Sizes: Multi core cables from 0.50 to 6 sq. mm for voltage grade upto 1.1 Kv.

Conductor: Copper – Flexible or extra flexible – Bare or Tinned.

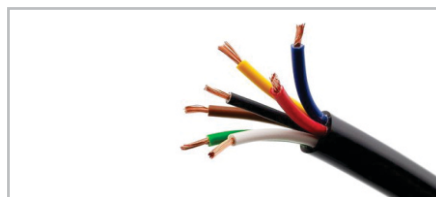
Insulation: PVC-GP/HR/FR/FRLS, Zero Halogen.

Inner Sheath: PVC-HR/FR/FRLS, Zero Halogen.

Outer Sheath: PVC-HR/FR/FRLS, Zero Halogen.

Specification: IS: 694, BS: 6500 and meeting customer requirements.

MULTICORE FLEXIBLE CABLES



Other Products:

1. Overhead Conductors (AAC, ACSR, AAAC, ACAR, ACSR/AW, AACSR, AACSR/AW, GAP, INVAR etc.)
2. Aerial Bunched Cables upto 120 sq mm.
3. Other specialised cables

MANUFACTURING FACILITIES



OUR VALUED CUSTOMERS



OTHER IMPORTANT CUSTOMERS

- ACME Telepower Ltd.
- Anupam Industries Ltd.
- Bhabha Atomic Research Centre
- Blue Star Ltd.
- Cabinet Secretariat, Govt. of India
- Damodar Valley Corporation
- DESEIN Ltd.
- Gas Authority of India Ltd.
- Honeywell Ltd.
- Indo Gulf Fertilizers & Chemical Corporation Ltd.
- Kanpur Fertilizers Ltd.
- M.N. Dastur & Co. Ltd.
- Mahanadi Coalfields Limited
- McNally Bharat Ltd.
- Miraj Instrumentation Services
- National Hydro Power Corporation Ltd.
- Shapoorji Pallonji & Co. Ltd.
- State Electricity Boards
- Sterling Electro Enterprises Pvt. Ltd.
- Tata Solar Ltd.
- Tata Projects Ltd.
- Voltas Ltd.
- Electricidade de Moçambique, Mozambique
- Paras Project Ventures, Nigeria
- Ministry of Energy, D.R. Congo
- Tendaho Sugar, Ethiopia
- UMEME, Uganda
- Punatsangchhu, Bhutan
- Maitree Thermal Power Corporation, Bangladesh
- Dhaka Mass Transit Co. Ltd., Bangladesh

BUSINESS SECTORS

- Power Generation, Transmission & Distribution
- Steel Plants
- Cement Plants
- Mining Industry
- Oil & Gas Refineries
- Automation Industry
- Renewable Energy
- Government & Public Sector Undertakings
- Chemical & Fertilizer Sector
- Non Ferrous Metal Smelters
- Military & Defence
- Railway & Metro Industry
- Crane Manufacturers
- Material Handling & Utility Vehicles
- Construction & Infrastructure
- EPC Contractors & Consultants



SPECIAL CABLES PVT. LTD.

(An ISO 9001: 2015, ISO 14001: 2015, ISO 45001: 2018 Company)

Corporate Office:

B-II/12, Mohan Co-operative Industrial Estate
Badarpur, New Delhi-110044, India
Phone : +91-11-29891011, 29893640
Email : sales@specialcables.co.in
Web : www.specialcables.co.in
CIN: U74899DL1982PTC013639

Manufacturing Unit: Plot No. 60-65, Sector-3
SIDCUL – IIE Pantnagar, Rudrapur- 263153
Dist.: U.S.Nagar, Uttarakhand, India
Phone : +91-594-4250781
E-mail : worksrdr@specialcables.co.in

BRANCH OFFICES & REPRESENTATIVES

Bangalore | Chennai | Hyderabad | Jamshedpur | Kolkata | Mumbai | Pune